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GPU Architecture

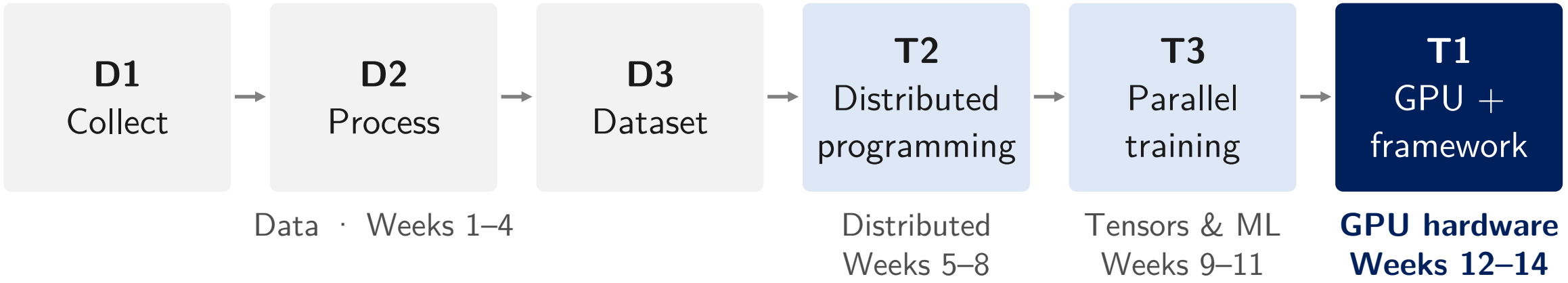
Throughput processors: SMs, warps, blocks, and grids

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COMP 329 · Spring 2027 · Week 12, Lecture 1

Where we are in the pipeline

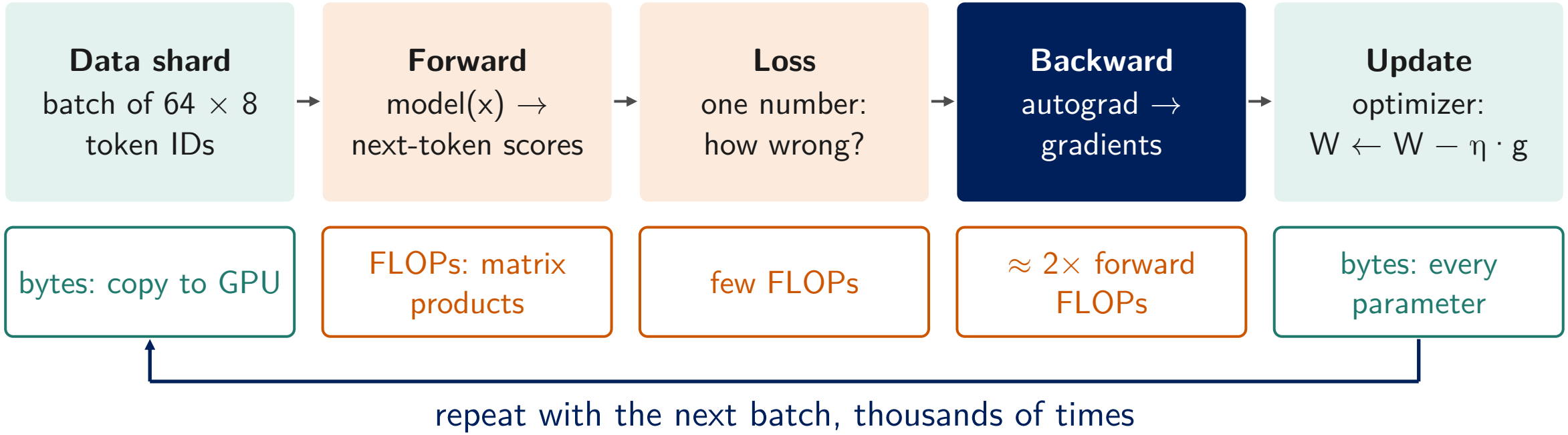


In our pipeline

Today: what happens inside **one GPU** running **one kernel** — the foundation for T1 and Lab 3.

Next: one training step again, and where the GPU fits among Weeks 5–14.

What one training step does



Key idea

Training time = number of steps × time per step. Weeks 5–11 split it across **many GPUs**; Weeks 12–14 make **each GPU fast**.

Weeks 5–14 at a glance

Problem		System component	Where
One GPU is too slow or too small	→	**Processes + communication: MPI, NCCL**	Wk 5–8 · T2
How do we split the work?	→	**Data, tensor, pipeline parallelism**	Wk 9–11 · T3
One step is millions of multiply-adds	→	**GPUs and kernels**	Wk 12–14 · T1

dark: today

Why do we need GPUs?

The job: train GPT-3 (175 billion parameters)
$\approx$ **3.14 $\times$ 10^{23}** floating-point operations, almost all multiply-adds

If one CPU core did all of this work, how long would it take?

One **CPU core**, at its peak
$\approx$ **52,000 years**

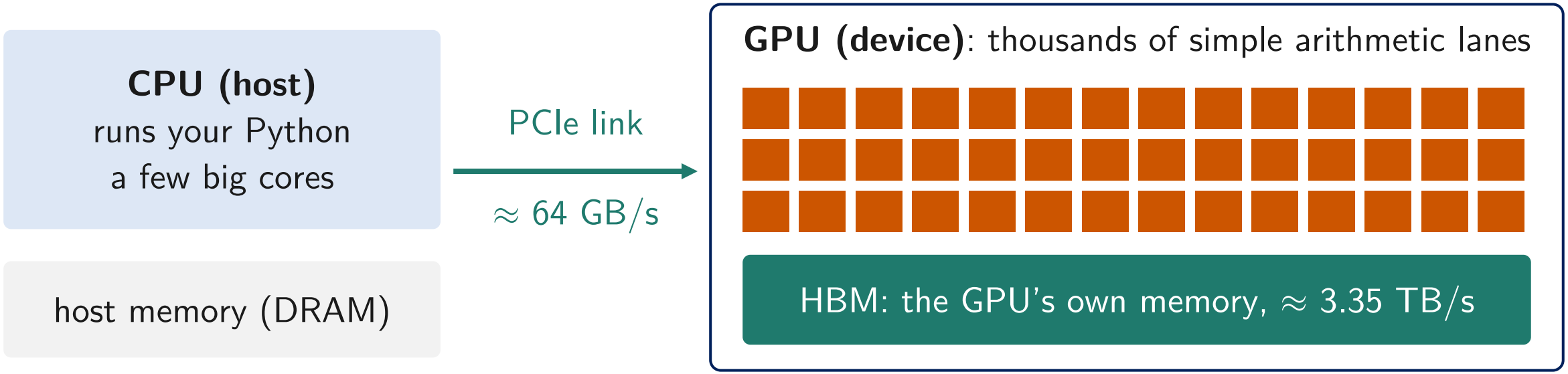
One **H100 GPU**, at its peak
$\approx$ **10 years**

Key idea

A GPU is **thousands of times** faster than a CPU core at this work — and even one GPU is too slow. So: many GPUs in T2 and T3, and fast GPUs in T1.

Source: Brown et al., NeurIPS 2020 (GPT-3 compute); CPU: assumed 3 GHz core, 2 AVX-512 FMA units; H100 datasheet (SXM5), BF16 peak.

What a GPU is, in plain words



1. Copy the data to HBM once, and keep it there.
2. Launch kernels: "run $z = x + y$ on all 10^8 elements."

Key idea

A **kernel**: one function the GPU runs on many elements at once. The CPU **launches** it.

Source: H100 datasheet (SXM5), HBM bandwidth; PCIe 5.0 x16, per direction.

Where GPUs show up

- **Graphics first:** every pixel is independent, so the chip is many simple cores.
- **2007, CUDA:** those cores become programmable for any arithmetic.
- **2012, AlexNet:** trained on two GPUs. Llama 3 used up to **16,384** H100s.
- **In PyTorch:** `x.to("cuda")`; every op on x then runs as GPU kernels.

The price

Fast only with **lots of independent work**

Data must fit in **80 GB** of GPU memory

Each launch costs CPU time: tiny jobs are **overhead-bound**

In our pipeline

Why GPUs? Training is a huge number of independent multiply-adds, and a GPU runs tens of thousands at once. In T1 and Lab 3 your kernels run on one.

Source: Krizhevsky et al., NeurIPS 2012; Llama Team, "The Llama 3 Herd of Models," 2024; H100 datasheet (SXM5).

Learning goals for today

- Explain why a GPU trades **latency** for **throughput**, and what Amdahl's law (Week 5) says it cannot fix.
- Name the parts of an **SM**: warp schedulers, registers, **CUDA cores**, **tensor cores**.
- Map **thread** $\rightarrow$ **warp** $\rightarrow$ **block** $\rightarrow$ **grid** onto the SMs, and count blocks, warps, and **waves**.
- Explain **SIMT** and predict the cost of **warp divergence**.

A puzzle to start

KERNEL 1

```
i = thread_index()
if i % 2 == 0:
    y[i] = slow_a(x[i])
else:
    y[i] = slow_b(x[i])
```

KERNEL 2

```
i = thread_index()
if (i // 32) % 2 == 0:
    y[i] = slow_a(x[i])
else:
    y[i] = slow_b(x[i])
```

Same n, half the elements in each branch, both branches equally slow. Which kernel is faster, and by how much?

Key idea

Kernel 2, about 2× faster. Threads run in groups of 32: in kernel 1 every group runs **both** branches; in kernel 2 each group runs **one**.

Part 1

From CPU to GPU

Why build a processor with thousands of slow, simple cores?

Latency versus throughput

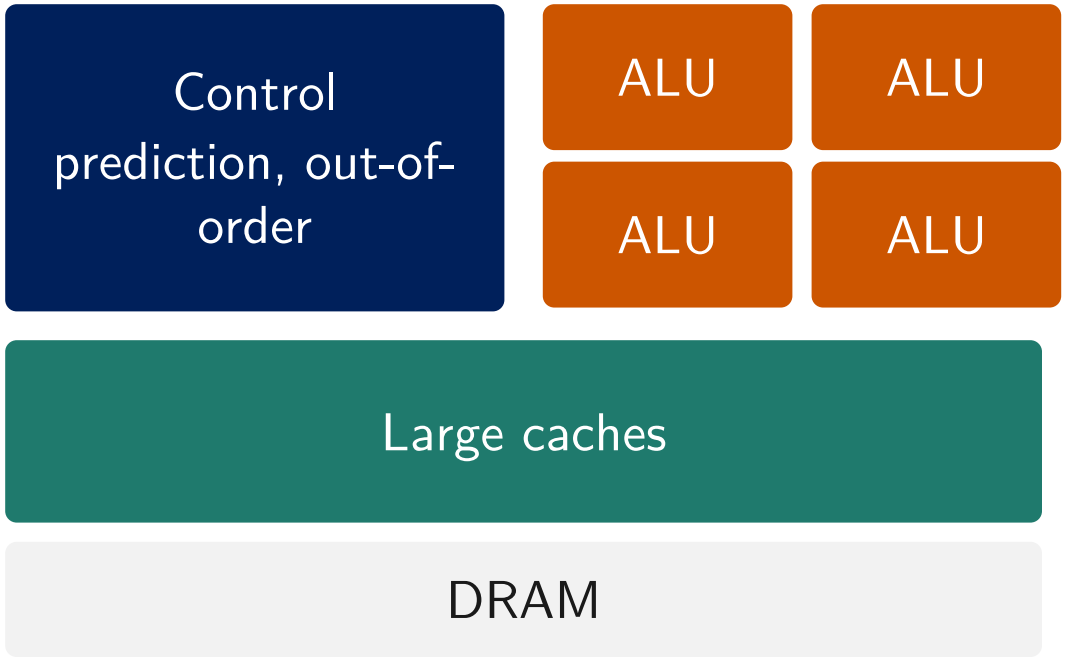
	Latency-oriented (CPU)	**Throughput-oriented (GPU)**
Goal	Finish **one** task fast	Finish **many** tasks per second
Cores	Tens of large cores	Thousands of simple lanes
Chip area	Caches, prediction, control	Arithmetic and registers
Hides slow memory by	Big caches, speculation	Switching to other threads

Key idea

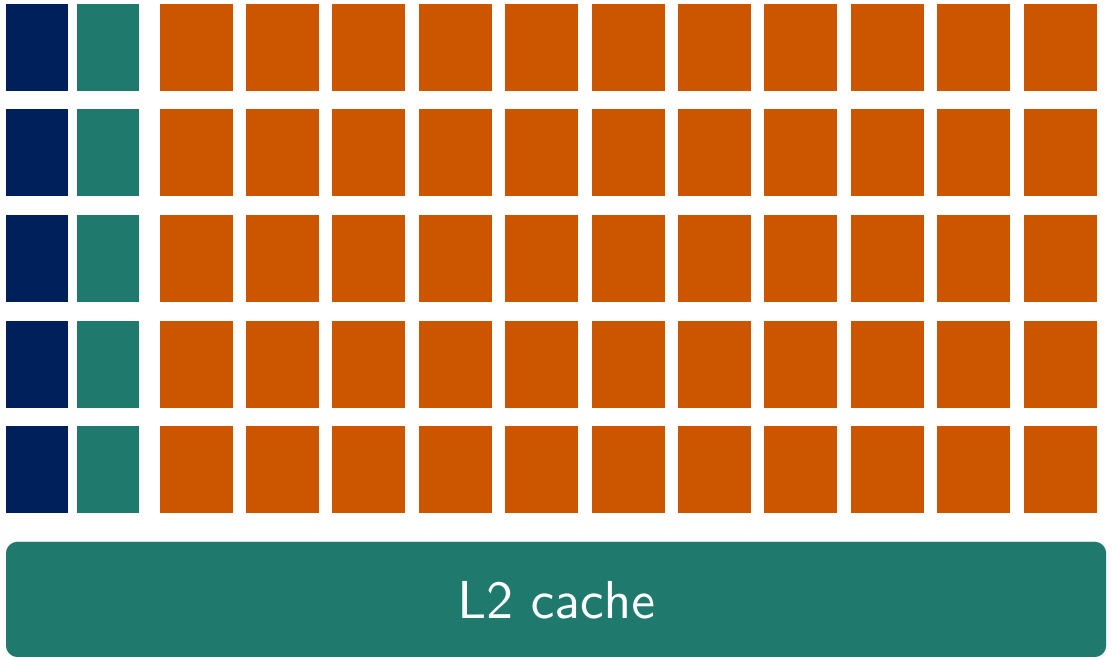
A sports car versus a fleet of buses. Training is a **throughput** problem.

Where the transistors go

CPU: a few powerful cores



GPU: many simple lanes

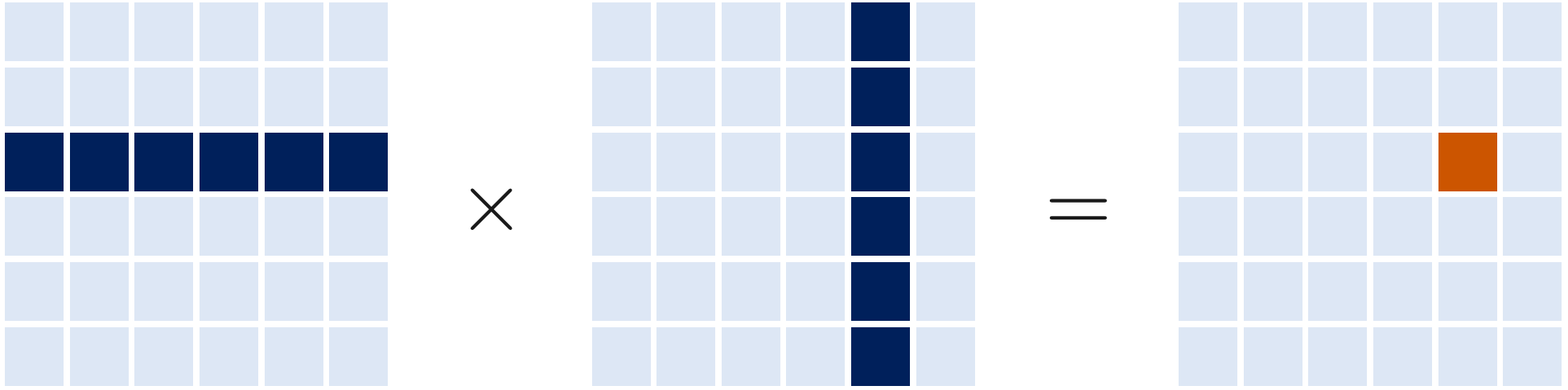


blue: control · green: cache · orange: arithmetic

Key idea

The GPU spends its area on arithmetic, so it needs **many threads** in flight to stay busy.

Why training fits the GPU



A row i × **B** column j = **C** C[i, j]

Key idea

Every C[i, j] is **independent**: $4096^2 \approx$ **16.8 million** tasks that can run at once.

Recall Amdahl's law (Week 5), on a GPU

$$\text{speedup} = 1 \;/\; (\; (1 - p) + p \;/\; s \;)$$

$$s \to \infty: \quad \text{speedup} \to 1 \;/\; (1 - p)$$

p = 0.95 on the GPU
5% left on the CPU
at most 20×

Watch out

The CPU side of a step — Python, data loading, **launching kernels** — bounds what any GPU can do. Lecture 2 measures the launch cost.

Check yourself

Two minutes with a neighbour.

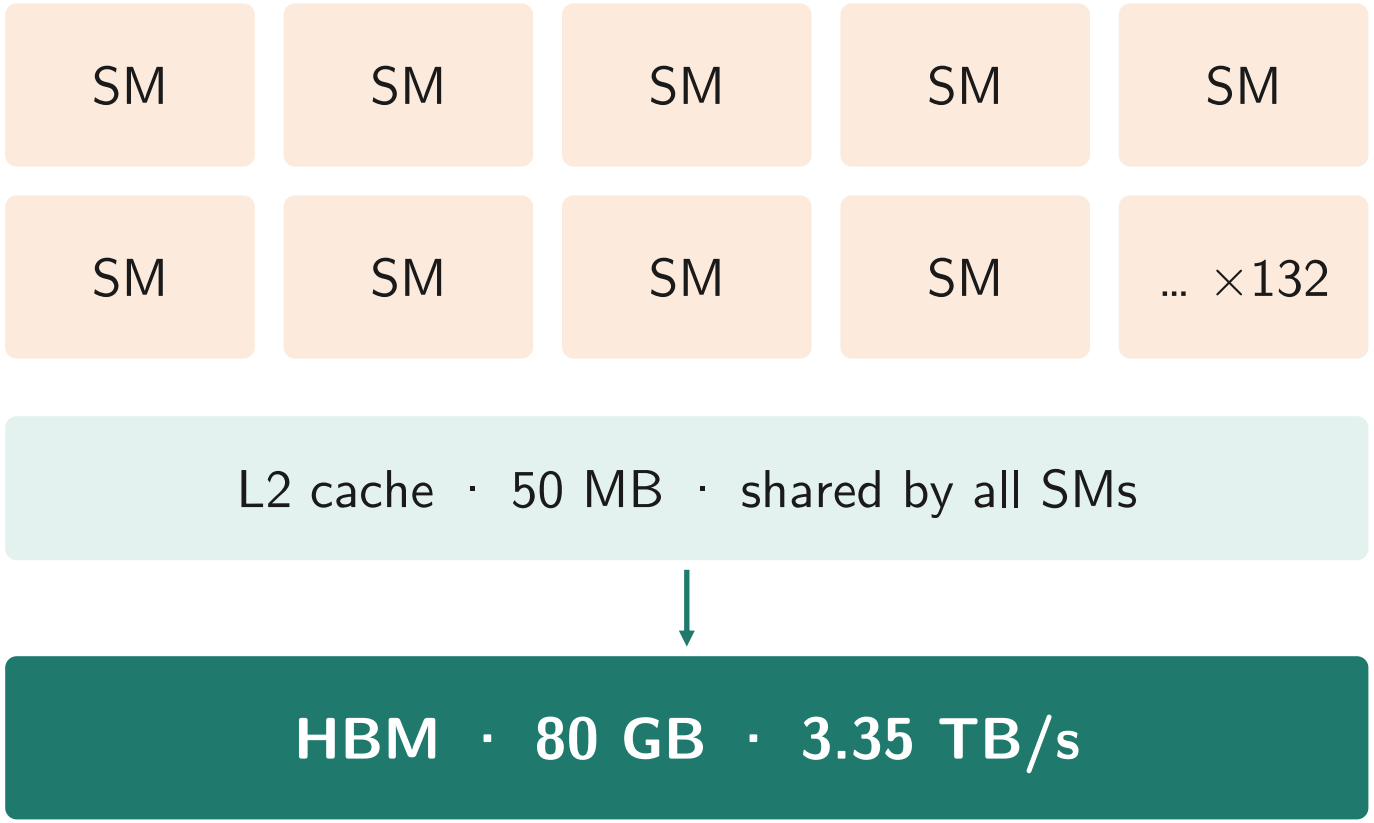
- Is training a latency problem or a throughput problem? Why?
- Name a computation that is hard to parallelize on a GPU, and say why.
- 80% of a program moves to a GPU that runs that part 100× faster. What is the overall speedup?

Part 2

Inside the GPU

What is a GPU made of?

A GPU is a grid of streaming multiprocessors



$$\begin{aligned}
&\text{Peak FP32} \\
&= \text{SMs} \times \text{lanes} \times 2 \times \text{clock} \\
&= 132 \times 128 \times 2 \times 1.98\ \text{GHz} \\
&\approx 67\ \text{TFLOP/s}
\end{aligned}$$

A fused multiply-add $a \cdot b + c$ counts as 2 FLOPs.

Source: NVIDIA H100 Tensor Core GPU datasheet (SXM5); NVIDIA H100 Tensor Core GPU Architecture whitepaper (2022).

Inside one SM

One of 4 sub-partitions

Warp scheduler	→ Picks a ready warp and issues **one instruction** each cycle.
Registers · 64 KB	→ Private, fastest storage: each thread has its own.
32 CUDA cores (FP32)	→ Execute that instruction for **32 threads at once**.
Tensor core	→ Multiplies small matrix tiles in one step.

× 4 sub-partitions share 256 KB of L1 / shared memory (up to 228 KB shared)

Source: NVIDIA H100 Tensor Core GPU Architecture whitepaper (2022), SM description.

CUDA cores and tensor cores

CUDA cores

128 per SM · 16,896 in all

one multiply-add per lane per cycle

any arithmetic: adds, tanh, compares

FP32 peak ≈ **67 TFLOP/s**

Tensor cores

4 per SM · 528 in all

one small matrix product per instruction

only matrix products

BF16 peak ≈ **989 TFLOP/s**

$$989 \;/\; 67 \approx 15\times$$

Key idea

Matrix products run ≈ **15×** faster than other arithmetic. Number formats and tile shapes: Week 14.

Source: H100 datasheet (SXM5), dense peaks (sparse ÷ 2); H100 whitepaper (4 tensor cores per SM).

Check yourself

Two minutes with a neighbour.

- Which work can run on tensor cores, and which cannot?
- Where does the 67 TFLOP/s FP32 peak come from?
- 4 schedulers, each issuing one 32-lane multiply-add per cycle: how many FLOPs per cycle per SM?

Part 3

Threads, warps, blocks, and grids

How does one program become 100,000 threads?

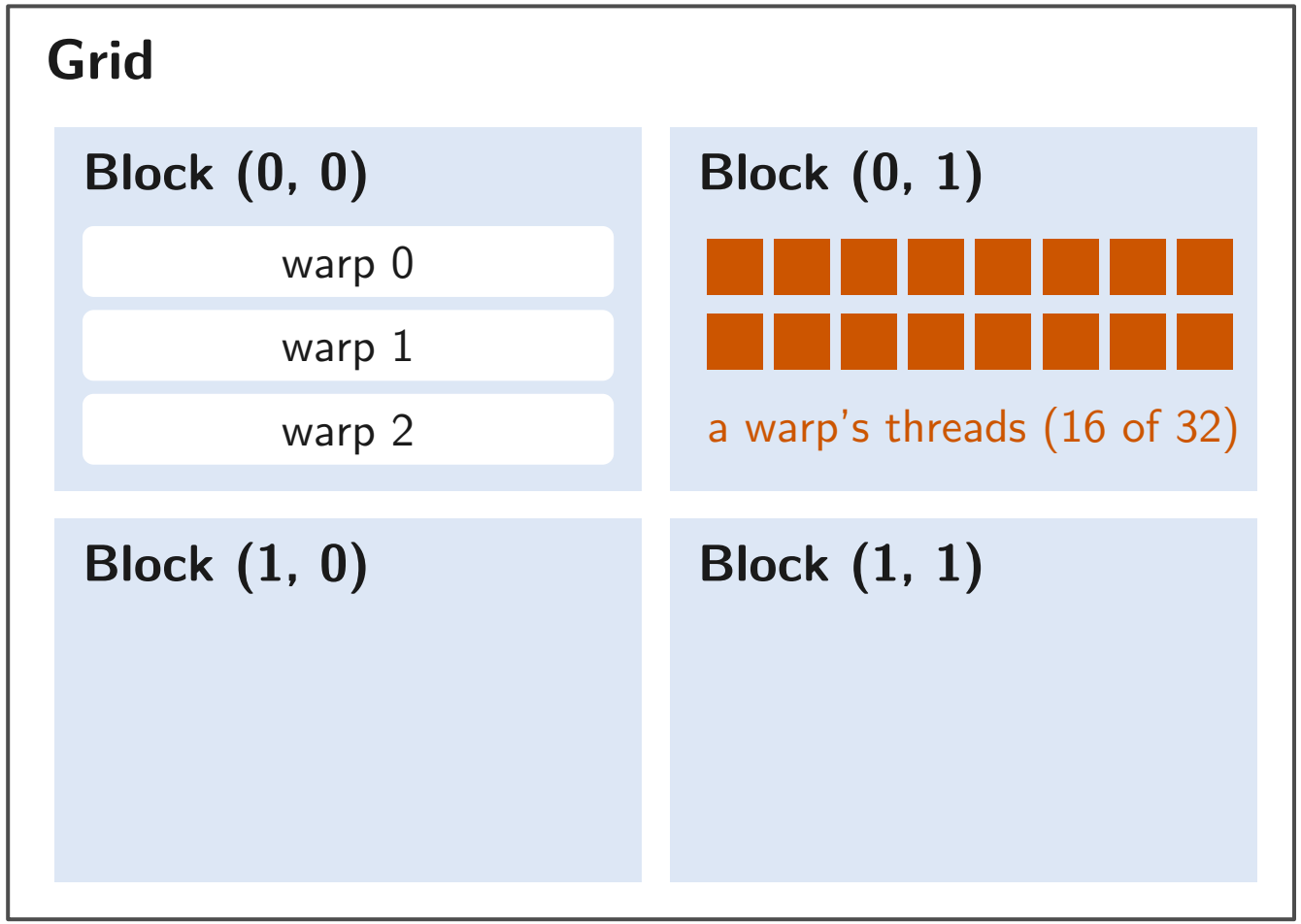
A GPU thread is not a Week 5 thread

	OS thread (Week 5)	GPU thread
How many	a handful per program	**millions** per launch
Created by	the operating system	the hardware, at launch
Switching	OS scheduler, microseconds	warp scheduler, **free**
Runs	on its own	in lockstep with **31 others**

Key idea

A GPU thread is **cheap**: one instance of the kernel with its own registers and index.

Software view: grid → block → warp → thread



Grid — every block of one launch; blocks must be **independent**.

Block — up to 1024 threads on **one SM**; can cooperate.

Warp — 32 threads that the hardware runs **together**.

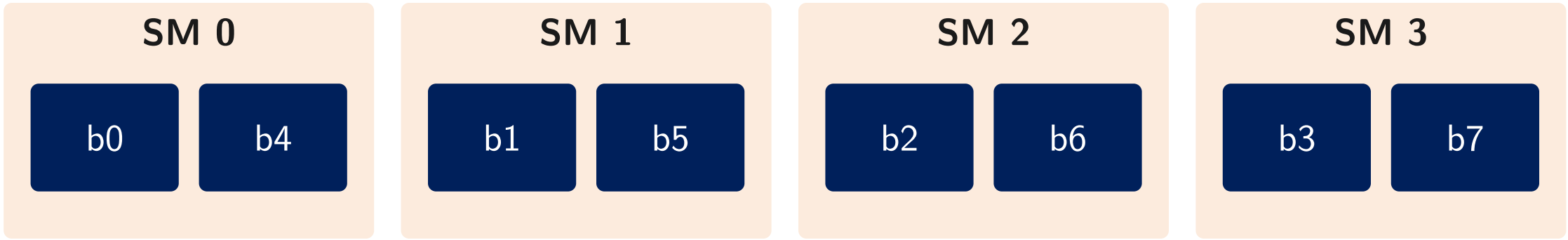
Thread — one instance of the kernel, with its own registers.

Source: NVIDIA CUDA C++ Programming Guide: warp size 32; at most 1024 threads per block.

Blocks are dealt out to SMs

b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 b11

the grid: 12 blocks



waiting: b8 b9 b10 b11 each starts when a slot frees up

Key idea

A block runs on **one** SM, start to finish, in **any** order — so blocks must be independent.

Source: NVIDIA CUDA C++ Programming Guide, technical specifications for compute capability 9.0 (H100).

Big grids run in waves

$$\text{per SM:}\quad 2{,}048 \;/\; 256 = 8 \text{ blocks}$$

$$\text{one wave:}\quad 132 \times 8 = 1{,}056 \text{ blocks}$$

$$n = 10^8\text{:}\quad 390{,}625 \text{ blocks} \approx 370 \text{ waves}$$

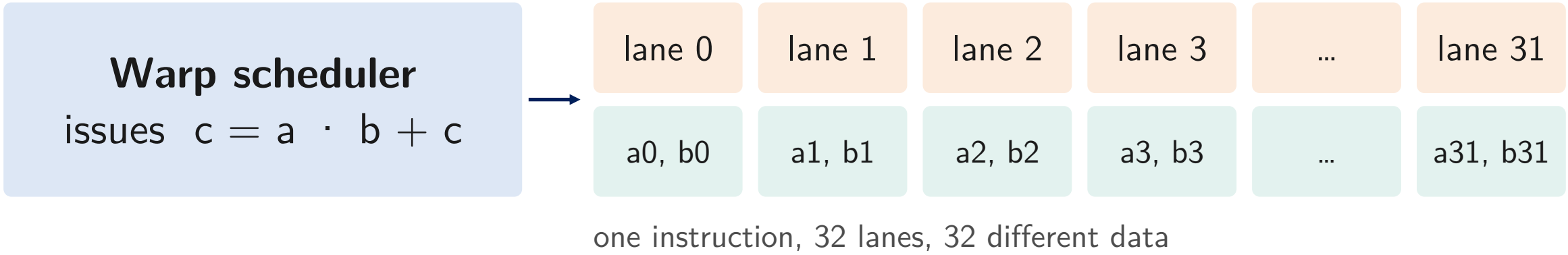
1,056 blocks $\rightarrow$ 1 wave
1,057 blocks $\rightarrow$ 2 waves, the 2nd with one block

Watch out

Many waves: the last partial wave hardly matters. **One or two** waves: it can double the time.

Source: NVIDIA CUDA C++ Programming Guide, technical specifications for compute capability 9.0 (H100). Assumes threads are the only limit.

SIMT: single instruction, multiple threads



- One fetch and decode drives **32** operations $\rightarrow$ cheap control.
- Each thread has **its own registers** and index; code is written for one thread.
- A warp computes $(f(x_0), f(x_1), \ldots, f(x_{31}))$ with one instruction.

Branch divergence

```
i = thread_index()
if x[i] > 0:          # lanes 0-15
    y[i] = slow_a(x[i])
else:                 # lanes 16-31
    y[i] = slow_b(x[i])
```

Pass 1: lanes run branch A



Pass 2: lanes run branch B



orange: active lane · grey: masked off

$$\text{efficiency} = \text{active lanes} \;/\; 32$$

$$50/50 \text{ split:}$$

$$\text{time} \approx \mathcal{T}(A) + \mathcal{T}(B)$$

Watch out

Divergence costs time only **within a warp**; different warps may take different paths freely.

Whether a warp diverges depends on the data

Condition, and the data	Warps that diverge	Time per warp
constant on runs of 32	none	T(A) or T(B)
`x[i] > 0`, random signs	almost all	T(A) + T(B)
`x[i] > 0`, x sorted	at most one	T(A) or T(B)

$$\text{random signs:}\quad \mathcal{P}(\text{all 32 agree}) = 2 \cdot 0.5^{32} \approx 4.7 \times 10^{-10}$$

Key idea

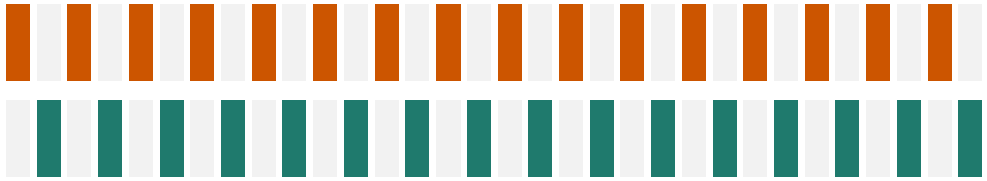
Arrange the data so that **neighbouring threads agree**; sorting or partitioning removes divergence.

Back to the puzzle

Kernel 1: `i % 2 == 0`

warp 0, pass A

warp 0, pass B



Kernel 2: `(i // 32) % 2 == 0`

warp 0, A only

warp 1, B only



$$\text{kernel 1:}\quad \mathcal{T}(A) + \mathcal{T}(B)$$

$$\text{kernel 2:}\quad (\mathcal{T}(A) + \mathcal{T}(B)) \,/\, 2$$

$$\text{per warp} \;\Rightarrow\; 2\times \text{ faster}$$

Key idea

Same work, same branches: what matters is whether the **32 threads of one warp** agree.

Check yourself

Two minutes with a neighbour.

- n = 5000, 256 threads per block: how many blocks, warps, and idle threads?
- Can block 7 safely read a value that block 3 writes in the same launch?
- `if x[i] > 0` on a **sorted** `x`: how many warps diverge?

IN-CLASS BUILD

Map a grid, cost a branch

Work in pairs, on paper, about eight minutes.

Exercise: map a grid, cost a branch

- $n = 10^6$, 256 threads per block: how many **blocks**, **warps**, and **idle threads**?
- How many blocks run at once on 132 SMs? How many **waves**?
- `if i % 4 == 0: A() else: B()`, A = 100 cycles, B = 20 cycles. Cycles per warp? Fraction of lane-cycles that do useful work?
- Rewrite the condition so that no warp diverges. What is the speedup?

USE

132 SMs $\cdot$ $\leq$ 2,048 threads per SM $\cdot$ warp = 32 threads

Exercise: solutions

Question	Answer
1. Blocks, warps, idle	3,907 blocks · 31,256 warps · 192 idle threads
2. At once, waves	$8 \times 132 = 1{,}056$ blocks · 4 waves (last: 739)
3. Divergent branch	$100 + 20 = 120$ cycles · $1{,}280 / 3{,}840 = 33\%$ useful
4. `(i // 32) % 4 == 0`	$\frac{1}{4} \cdot 100 + \frac{3}{4} \cdot 20 = 40$ cycles per warp $\rightarrow$ **3×**

Key idea

Count with ceilings; ask what the **32 threads of one warp** do. Lecture 2 adds what the warps **wait** for.

Key takeaways

- GPUs trade single-thread **latency** for **throughput**; training is a throughput problem.
- Amdahl (Week 5): the CPU side of a step bounds what any GPU can do.
- An H100 is **132 SMs**: **CUDA cores** for any arithmetic, **tensor cores** for matrix products.
- **Grid** $\rightarrow$ **block** $\rightarrow$ **warp** $\rightarrow$ **thread**: a block lives on one SM; big grids run in **waves**.
- **SIMT**: one instruction per warp of 32; **divergence** within a warp runs both paths.

Next time, and readings

When	What
Lecture 2 (this week)	Execution model: latency hiding, occupancy, launches, copies
Friday (this week)	T3 due
After Lecture 2	T1 and Lab 3 released
Week 13	The memory hierarchy; arithmetic intensity and the roofline

READ BEFORE LECTURE 2

CUDA C++ Programming Guide: *Hardware Implementation* (SIMT, warps).

PyTorch documentation: *CUDA semantics* (asynchronous execution).